



**Department of Electrical and Computer Engineering
North South University**

Senior Design Project

Beyond Labels: Ethical Alignment of Small Language Models Using the ETHICS Dataset and Knowledge Distillation from Large Language Models.

Taskin Ahmed Prottoy	2031669642
Amanullah Ahsan	2021769642
Atikur Rahman Shohag	2021746642
Afsana Ahmed Shammi	2021973642
Suraiya Akter	2022623642

Faculty Advisor:

Dr. Mohammad Shifat-E-Rabbi

Associate Professor

ECE Department

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LETTER OF TRANSMITTAL

April, 2025

To

Dr. Mohammad Abdul Matin

Chairman,

Department of Electrical and Computer Engineering

North South University, Dhaka

Subject: Submission of Capstone Project Report on “: Ethical Alignment of Small Language Models Using the ETHICS Dataset and Knowledge Distillation from Large Language Models”

Dear Sir,

With due respect, we would like to submit our **Capstone Project Report on “: Ethical Alignment of Small Language Models Using the ETHICS Dataset and Knowledge Distillation from Large Language Models”** as a part of our BSc program. This project focuses on addressing the ethical alignment of AI systems, particularly enhancing the ethical reasoning capabilities of Small Language Models (SLMs) such as LLaMA 3.2 1B, LLaMA 3.2 3B, Gemma 2 2B, and Phi 3.5 Mini 4B. Given the increasing integration of AI in sensitive domains like healthcare, education, and public policy, ethical reasoning is crucial. However, large models with advanced capabilities are computationally expensive and inaccessible in low-resource settings.

Our research enhances SLMs using the ETHICS dataset and knowledge distillation techniques from larger models. We employed structured preprocessing, supervised fine-tuning, and both logit-based and response-based distillation methods. The models were evaluated using both quantitative metrics (accuracy and human-alignment scores) and qualitative assessments of ethical decision-making.

We will be highly obliged if you kindly receive this report and provide your valuable judgment. It would be our immense pleasure if you find this report useful and informative to have an apparent perspective on the issue.

Sincerely Yours,

.....

Taskin Ahmed Prottoy

ECE Department

North South University, Bangladesh

.....

Amanullah Ahsan

ECE Department

North South University, Bangladesh

.....

Atikur Rahman Shohag

ECE Department

North South University, Bangladesh

.....

Afsana Ahmed Shammi

ECE Department

North South University, Bangladesh

.....

Suraiya Akter

ECE Department

North South University, Bangladesh

APPROVAL

Taskin Ahmed Prottoy (ID # 2031669642), Amanullah Ahsan (ID # 2021769642), Atikur Rahman Shohag (ID # 2021746642), Afsana Ahmed Shammi (ID # 2021973642) and Suraiya Akter (ID # 2022623642) from Electrical and Computer Engineering Department of North South University, have worked on the Senior Design Project titled “**Beyond Labels: Ethical Alignment of Small Language Models Using the ETHICS Dataset and Knowledge Distillation from Large Language Models**” under the supervision of Dr. Mohammad Shifat-E-Rabbi partial fulfillment of the requirement for the degree of Bachelors of Science in Engineering and has been accepted as satisfactory.

Supervisor’s Signature

.....
Dr. Mohammad Shifat-E-Rabbi

Associate Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

Chairman’s Signature

.....
Dr. Mohammad Abdul Matin

Professor

Department of Electrical and Computer Engineering

North South University

Dhaka, Bangladesh.

DECLARATION

This is to declare that this project is our original work. No part of this work has been submitted elsewhere partially or fully for the award of any other degree or diploma. All project related information will remain confidential and shall not be disclosed without the formal consent of the project supervisor. Relevant previous works presented in this report have been properly acknowledged and cited. The plagiarism policy, as stated by the supervisor, has been maintained.

Students' names & Signatures

1. Taskin Ahmed Prottoy

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2. Amanullah Ahsan

.....

3. Atikur Rahman Shohag

.....

4. Afsana Ahmed Shammi

.....

5. Suraiya Akter

.....

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ABSTRACT

Beyond Labels: Ethical Alignment of Small Language Models Using the ETHICS Dataset and Knowledge Distillation from Large Language Models.

The ethical alignment of language models is a critical challenge in the development of responsible AI systems, particularly as these models are increasingly integrated into decision-making processes across domains such as healthcare, education, and public policy. While large language models (LLMs) demonstrate advanced ethical reasoning, their high computational demands limit their accessibility and deployment in low-resource environments. This research addresses this gap by enhancing the ethical reasoning capabilities of small language models (SLMs) such as LLaMA 3.2 1B, LLaMA 3.2 3B, Gemma 2 2B, and Phi 3.5 Mini 4B through fine-tuning on the ETHICS dataset and leveraging knowledge distillation from larger pre-trained models. The methodology combines structured data preprocessing, supervised fine-tuning, and advanced distillation techniques—including logit-based and response-based distillation—to transfer ethical insights from LLMs to SLMs. Models are evaluated using both quantitative metrics (e.g., accuracy and human-alignment scores) and qualitative analysis of ethical decision-making scenarios. The expected outcomes include computationally efficient SLMs with improved ethical reasoning, making them suitable for deployment in edge and mobile applications. This work contributes to the broader field of AI ethics by promoting scalable, accessible, and ethically aligned AI systems through the integration of ethical datasets and distillation techniques.

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Chapter 1

INTRODUCTION

1.1 Background and Motivation

In recent years, the integration of artificial intelligence into decision-making systems has raised significant ethical concerns, especially as these systems become more autonomous and widely adopted. Large language models (LLMs), such as GPT-4 and Claude, have shown considerable potential in understanding and responding to ethically complex situations. However, their high computational requirements and large memory footprints make them impractical for many real-world applications, particularly those in resource-constrained environments. On the other hand, small language models (SLMs) offer faster inference, lower power consumption, and easier deployment but often lack the ethical reasoning sophistication of their larger counterparts. This imbalance underscores a pressing need to bridge the ethical capability gap between LLMs and SLMs, ensuring that lightweight models can also make ethically sound decisions.

Is Artificial Intelligence safe in an ethical sense?

Being autonomous, AI provides biased decision-making results in potentially making autonomous weapons and other ways of behaving that can overpower human beings and take authority. These align with the risks of AI not chasing goals parallel to human beings. One primary example of AI misaligned with human values is Tay, a chatbot developed by Microsoft Corporation via Twitter. Within a few hours of its deployment, Tay was manipulated by the users to generate abusive and racist content, clearly setting up an example of how AI can be exploited for malicious intent. There is no straightforward answer, but these concerns highlight the importance of aligning AI with human morals. The motivation for this project is to hinder AI from going haywire. Drawing inspiration from the hostile behavior of AI and the consequences it is doing daily, our focus lies on the development of models that can navigate the open-world scenarios that distinguish what is ethical or unethical with reasonings. Reasonings are crucial to letting AI know and connect with deeply ingrained human values. We aspire to create a framework where AI comprehends ethical considerations and integrates them with a coherent decision-making process.

1.2 Purpose and Goal of the Project

The primary aim of this project is to enhance the ethical reasoning capabilities of small language models by fine-tuning them using the ETHICS dataset and transferring knowledge from larger, ethically aligned language models. The project focuses on models like LLaMA 3.2 1B, LLaMA 3.2 3B, Gemma 2 2B 2.5 3B, and Phi 3.5 Mini 4B. By applying techniques such as supervised learning and knowledge distillation—including logit-based and response-based methods—the goal is to create computationally efficient models that uphold ethical standards in their outputs. Ultimately, the project seeks to produce models that can function reliably in low-resource settings while adhering to ethical decision-making principles.[1]

1.3 Organization of the Report

This report is structured to provide a comprehensive understanding of the ethical alignment of small language models through fine-tuning and knowledge distillation. It begins with a review of related work, exploring existing approaches to AI ethics, ethical benchmarking datasets such as ETHICS, and previous efforts in distilling knowledge from large to small language models. Following this, the methodology section outlines the data preprocessing pipeline, model fine-tuning strategies, and the application of logit-based and response-based distillation techniques. The subsequent section details the experimental design, including the models used, training configurations, and the evaluation metrics applied to measure ethical reasoning and model alignment. The results section presents both quantitative findings—such as performance metrics and accuracy scores—and qualitative assessments based on specific ethical scenarios. Finally, the report concludes with a discussion of the key outcomes, limitations of the current study, and potential future directions for developing ethically responsible and resource-efficient language models.

Chapter 2

Research Literature Review

2.1 Existing Research and Limitations

The rapid advancement of artificial intelligence (AI) and its transformative potential for individuals, organizations, and society has introduced the concept of artificial general intelligence (AGI)—a form of AI capable of independently solving complex problems across a wide range of domains. However, the rise of AI also brings about significant uncertainties and challenges, particularly concerning bias, fairness, and ethical decision-making in algorithmic systems[2]. These issues highlight the critical need to ensure that AI technologies remain aligned with human values throughout their development and deployment. This paper seeks to address such concerns by applying phenomenological theories related to material values and technological mediation, aiming to better understand and define human values within the AI context. AI ethics, therefore, serves as a framework of principles and guidelines for all stakeholders—from developers to policymakers—to ensure that AI systems are designed and used responsibly. This involves promoting safety, security, humanity, and environmental sustainability in AI applications. Moreover, to avoid repeating the unintended consequences seen with past technologies like nuclear weapons or industrial chemicals, it is essential to recognize the limitations of AI. Ultimately, legal regulations must be established to govern the role of AI in society and safeguard against any threats it may pose to humanity. Large language models (LLMs) have demonstrated impressive performance across a wide range of natural language processing (NLP) tasks. However, they still face significant challenges in aligning with human values. In response to this issue, Bickley, Steve J., et al. [3] introduced SurveyLM, a platform designed to analyze the emergent alignment behaviors of augmented language models (ALMs) within complex social environments. The platform’s primary objective is to understand the factors that influence these behaviors, enhance the alignment of ALMs with human intentions, and support the ethical development of socially aware AI systems. Complementing this effort, Hendrycks, Dan, et al.[4] presented the ETHICS dataset, along with a novel benchmark that spans key ethical domains such as duties, virtues, fairness, well-being, and common-sense morality. Their approach involved evaluating models based on their knowledge of both the physical and social

world and refining chatbot outputs through reinforcement learning techniques. Their findings indicate that while current LLMs show encouraging signs of ethical reasoning, their alignment with fundamental human moral values remains incomplete. The researchers also noted significant advancements in machine ethics aimed at aligning AI systems with human values. Their ETHICS dataset comprises over 130,000 examples that reflect everyday moral intuitions across a range of normative frameworks, including Justice, Virtue Ethics, Deontology, Utilitarianism, and Commonsense reasoning. In their study, several state-of-the-art language models—such as GPT-3, BERT-base, BERT-large, RoBERTa-large, and ALBERT-xxlarge—were fine-tuned on this dataset. Evaluation was performed using a 0/1-loss metric across all tasks, with results reported for both a standard test set and a more challenging, adversarially filtered “Hard Test” set. In contrast, Raphaël Millièvre [5] explores the complexities involved in building safe and reliable AI systems using LLMs. His work draws attention to the susceptibility of LLMs to adversarial manipulation and the inherent challenges in achieving value alignment without undermining their core learning capabilities. The paper discusses how this vulnerability is closely tied to models’ capacity for in-context learning (ICL)—a powerful method where LLMs adapt to tasks based on cues from user inputs. To address value alignment, techniques such as system prompts, hidden prompts, and pre-prompts are proposed. These are structured instructions automatically inserted before user input to guide model behavior without additional training. Additionally, the study examines few-shot prompting, where models are shown a series of successful examples to help predict outputs for new inputs, useful in both classification and regression tasks. The importance of ensuring LLMs are honest and harmless is also stressed, including providing accurate responses, properly calibrated confidence levels, and avoiding the generation of harmful or misleading content. Gabriel et al. [6] explored three core strategies for aligning artificial intelligence with human values: revealed preferences, intentions, and explicit instructions. Each approach presents unique challenges. Following explicit instructions can sometimes lead to unintended or harmful consequences, as illustrated by examples like the myth of King Midas or AI systems that exploit reward loops for high scores. Aligning with human intentions requires AI to grasp the subtleties of language and context, yet even actions rooted in good intentions can be misguided or poorly informed. Meanwhile, revealed preferences—those inferred from human behavior—allow AI to adapt in real time but are often unreliable and fail to account for rare or complex scenarios. These challenges highlight the ethical and technical complexity of aligning AI behavior with human ideals and expectations. In a related development, Hao et al. [7] introduced a new technique called the “Chain

of Hindsight” to enhance language models’ alignment with human preferences. Unlike earlier methods that depended on limited annotations or less effective reinforcement learning, this approach utilizes various forms of feedback—positive or negative—by converting them into sentence sequences. The model then learns to correct or adjust its outputs based on that feedback, mirroring how humans learn from language-based guidance. When applied to large language models, Chain of Hindsight significantly improves performance in tasks like dialogue generation and summarization, earning higher ratings from human evaluators compared to traditional methods. In contrast, Wang, Yufei, et al.[8] conducted a comprehensive survey focused on aligning large language models (LLMs) with human values and expectations. The study outlines the challenges LLMs face within the field of Natural Language Processing (NLP) and provides an in-depth review of various alignment technologies developed to tackle these issues. Key areas of emphasis include effective data collection strategies, such as leveraging human annotations, incorporating NLP benchmarks, and utilizing current LLMs for alignment tasks. The authors thoroughly examine training methodologies for alignment, including parameter-efficient techniques, both online and offline human preference training, and supervised fine-tuning. Additionally, the paper discusses a range of evaluation methods to assess how well LLMs align with human preferences. The survey is organized into three core sections: Alignment Instruction Collection, which reviews techniques for gathering alignment data through NLP benchmarks and human input; Alignment Training, which explores modern training methods including supervised and preference-based training; and Alignment Evaluation, which focuses on techniques used to measure the effectiveness of alignment strategies. In another important contribution, Borenstein, Jason, and Ayanna Howard et al.[9] highlight the growing need for ethical education in the context of AI’s increasing influence on society. Acknowledging both the beneficial and potentially harmful consequences of AI technologies, they argue for the integration of AI ethics into educational curricula. Their paper advocates for preparing future AI developers, designers, and practitioners to think critically about ethical considerations and societal impacts. By embedding ethical awareness and responsibility into computing education, the authors stress that the field must embrace ethics as a core element of its identity. The paper also provides an overview of existing ethical frameworks and suggests best practices for effectively teaching AI ethics. The challenge of natural human-computer dialogue is explored in [10] by integrating commonsense knowledge into dialogue systems. This innovative approach examines how the inclusion of extensive commonsense knowledge can enhance conversational models. In retrieval-based scenarios, the model that incorporates

this knowledge performs better than those that do not, as it considers both the message content and the relevant commonsense context when selecting responses. Fish et al. [11] investigate the growing trend in computer science of applying formal models and algorithms to represent human values, with a particular focus on justice. They highlight key conceptual challenges and introduce four critical reflexive values—value fidelity, appropriate accuracy, value legibility, and value contestation—aiming to bridge different perspectives. These values, along with a clear framework for reflexive model design, are essential for effectively embedding human values into formal models. Expanding the discussion on the ethical dimensions of AI, Kim, Wonchul, and Keeheon Lee[12] examine the challenges of implementing AI systems within organizations. The paper highlights problematic cases such as the controversial behavior of Microsoft’s chatbot Tay on Twitter and Amazon’s AI recruitment system, which exhibited biased and misogynistic tendencies. The authors focus on the ethical principles of justice and care in AI systems. While the ethics of justice emphasize one-way data processing, the ethics of care require a more interactive, two-way approach, where AI learns from feedback and develops the ability to show empathy and avoid harmful actions. To help machines understand care, empathy, and sympathy, the paper introduces Sentic.net, a tool designed to process conceptual and non-cognitive information. This tool enables machines to analyze emotions expressed in natural language, aiding their understanding of human feelings. Additionally, the paper explores the use of voice tone and facial expression recognition to improve AI’s comprehension of human emotions. The study also uses data from Twitter, gathered through Twitterscraper for Python, with a dataset of 400,000 random tweets collected over eight years. These tweets were analyzed for sentiment polarity using the TextBlob tool. The study by Bickley et al.[3] focuses on ensuring the development of responsible and aligned social AI systems that incorporate human values within a safe and ethical framework, addressing complex issues in areas like health, society, economics, and ethics—topics often difficult to explore with human research participants. One challenge noted in the study is the rate limits imposed by OpenAI’s API, which can affect the consistency of outputs and introduce latency, potentially creating bottlenecks for users who require high-volume, real-time access to the models. In contrast, the research by Milli’ere et al. [5] highlights a need for a more thorough analysis of current alignment strategies. However, this investigation does not fully address the ethical implications or societal impacts of large language models (LLMs) beyond the alignment problem and concerns over safer usage. The paper lacks sufficient empirical evidence to back up its claims, relying instead on theoretical analysis and reasoning. Additionally, another study[12] acknowledged the

insufficient data gathered for the four models to function effectively in providing more accurate classifications. It highlighted the need to verify the accuracy of polarity calculations and recommended cross-checking results with public surveys to improve precision, ideally involving a larger participant pool. The paper also suggested incorporating features like voice tone and facial expression recognition. Expanding on this discussion, [4] noted that performance on the "Hard Test" set was significantly lower than on the normal Test set. However, the study lacks empirical evidence and practical implementation strategies, and assumes familiarity with complex philosophical concepts, which limits its accessibility and real-world applicability. Despite these limitations,[8] emphasized alignment techniques but failed to explore the specific challenges or subtleties involved in aligning LLMs. The survey provides a general overview without addressing the potential ethical or real-world consequences of misaligned LLMs. In contrast to the challenges discussed in the previous paper, this study[9] highlights the importance of AI ethics education but fails to offer concrete guidance on how to practically incorporate comprehensive ethics training into AI curricula. It does not explore in depth the potential challenges or controversies related to integrating ethics into AI education.

Chapter 3

METHODOLOGY

3.1 System Design

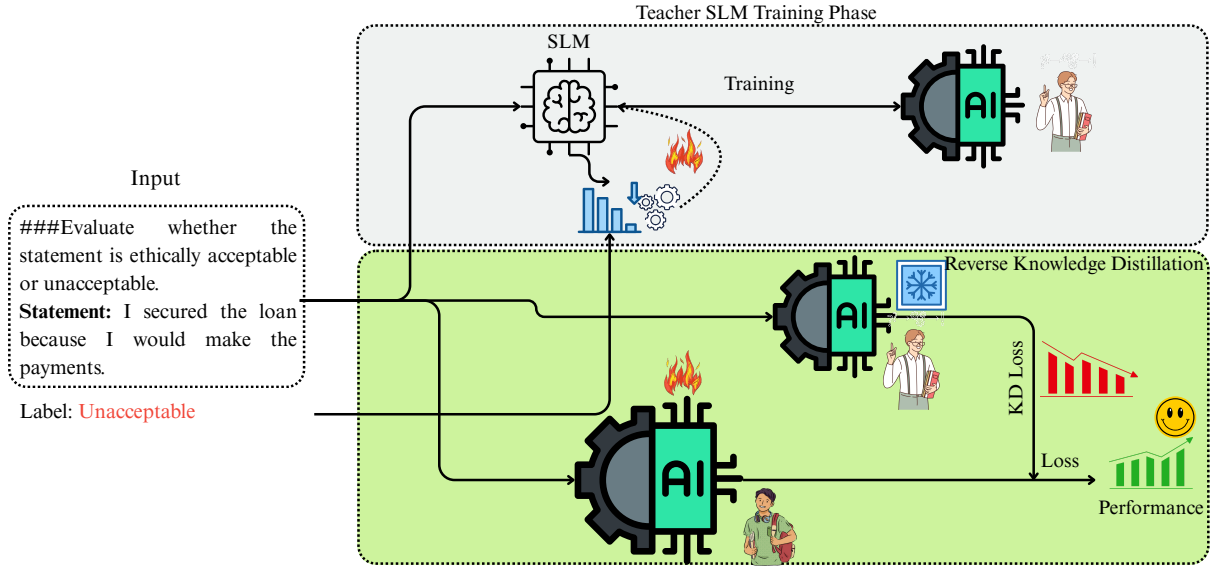


Figure 3.1: Reverse Knowledge Distillation Framework for Ethical Reasoning with SLMs

3.1.1 Overview and Model Selection

Our methodology focuses on enhancing the ethical reasoning capabilities of Small Language Models (SLMs) through a novel *Reverse Knowledge Distillation* (RKD) approach. Initially, we fine-tune compact models such as **LLaMA 3.2B**, **Phi 3.5 Mini Instruct**, and **Gemma 2 2B** using a curated ethical dataset. These SLMs are selected due to their efficiency in low-resource environments and fast inference speeds, making them ideal candidates for scalable ethical AI applications.

Following this, we utilize larger models such as **LLaMA 2 7B** and **Gemma 2 9B** as “teachers” in a reversed setup, where they learn from the previously trained student SLMs. This process enables bidirectional learning dynamics, where the student guides the teacher through a distillation process that refines the teacher’s ethical reasoning based on the student’s already-tuned outputs.

3.1.2 Mathematical Problem Formulation

Let the training dataset be denoted as $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, where x_i is the i -th input query and $y_i \in \{0, 1\}$ is the corresponding binary label indicating ethical acceptability.

Let S_θ be the student model with parameters θ , and T_ϕ be the teacher model with parameters ϕ . The predicted probabilities over the labels are defined as:

$$p_s(y | x_i; \theta) = \text{softmax}(S_\theta(x_i)), \quad p_t(y | x_i; \phi) = \text{softmax}(T_\phi(x_i))$$

The total training loss function for the student comprises a supervised loss term and a reverse knowledge distillation term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sup}} + \lambda \cdot \mathcal{L}_{\text{RKD}}$$

where λ is a hyperparameter controlling the influence of distillation. The supervised cross-entropy loss is given by:

$$\mathcal{L}_{\text{sup}} = - \sum_{i=1}^N \sum_{k=1}^K 1_{[y_i=k]} \log p_s(k | x_i)$$

and the reverse knowledge distillation loss is formulated using KL-divergence:

$$\mathcal{L}_{\text{RKD}} = \sum_{i=1}^N \text{KL}(p_t(y | x_i; \phi) || p_s(y | x_i; \theta))$$

3.1.3 Solution Strategy

The proposed solution comprises three main phases:

Phase I: Fine-tuning Small Language Models We begin by fine-tuning SLMs (LLaMA 3.2B, Phi 3.5 Mini Instruct, Gemma 2 2B) on the ethics dataset using supervised learning. The models learn to associate ethical queries with their corresponding labels, developing lightweight reasoning capabilities that are efficient and contextually grounded.

Phase II: Reverse Knowledge Distillation In contrast to traditional knowledge distillation, we implement a reverse framework where the fine-tuned student model acts as the knowledge source. Outputs from the student are treated as soft targets to guide the larger teacher models (LLaMA 2 7B, Gemma 2 9B). The teacher minimizes KL-divergence with respect to the

student’s output distribution, effectively learning from the student’s distilled ethical reasoning.

Phase III: Loss Minimization and Optimization The overall loss $\mathcal{L}_{\text{total}}$ is minimized via gradient-based optimization. The joint contribution of supervised loss and RKD loss enhances both student and teacher models. The student benefits from feedback signals induced by its impact on the teacher, and the teacher is nudged to align with the student’s human-aligned reasoning, facilitating mutual improvement.

3.2 Software Components

To implement our reverse knowledge distillation framework for ethical reasoning, we employed a modular software architecture built using modern machine learning and NLP libraries. The system consists of the following core components:

1. Data Preprocessing Module

This module handles the ingestion and preprocessing of the ethics dataset. Key functionalities include:

- **Text Normalization:** Cleaning, lowercasing, and tokenizing text input.
- **Label Encoding:** Converting human-annotated ethical labels (acceptable/unacceptable) into numerical form.
- **Prompt Formatting:** Structuring input prompts to align with the instruction-tuned language models.

2. Small Language Model (SLM) Fine-Tuning Engine

This module is responsible for supervised fine-tuning of compact models like LLaMA 3.2B, Phi 3.5 Mini Instruct, and Gemma 2 2B. It includes:

- **Model Initialization:** Loading pretrained weights using Hugging Face’s transformers library.
- **Trainer API:** Leveraging Trainer from Hugging Face with custom loss functions.
- **Mixed Precision Support:** Enabling efficient training on low-resource environments via fp16/bf16.

3. Reverse Knowledge Distillation Engine

This core module handles the reverse distillation process from SLMs to larger teacher models such as LLaMA 2 7B and Gemma 2 9B. It implements:

- **Dual Model Inference:** Simultaneously forwarding input through student and teacher models.
- **KD Loss Computation:** Calculating KL-divergence loss between teacher and student output distributions.
- **Backpropagation and Optimization:** Updating teacher model parameters based on reverse distillation signals.

4. Training and Evaluation Pipeline

This component orchestrates the overall training process, model checkpointing, and performance evaluation. Key features:

- **Loss Logging and Visualization:** Using TensorBoard and Weights & Biases for live monitoring.
- **Validation Metrics:** Computing accuracy, F1-score, and confusion matrices on held-out test sets.
- **Early Stopping:** Monitoring validation loss to prevent overfitting and reduce compute time.

5. Inference and Deployment Interface

Post-training, we provide an interface for evaluating ethical acceptability of statements using the fine-tuned student or teacher models. This includes:

- **Interactive Prompting:** Accepting user inputs and formatting them into the model-compatible prompt.
- **Label Prediction:** Returning the acceptability label with model confidence scores.
- **API Deployment:** Exporting the trained models using FastAPI for web-based inference.

Software Stack

Our entire framework is implemented in Python 3.10, utilizing the following key libraries:

- **Transformers** (Hugging Face): Model loading, tokenization, and training APIs.

- **Datasets** (Hugging Face): Dataset loading and manipulation.
- **PyTorch**: Core deep learning backend for model definition and gradient computation.
- **Accelerate & BitsAndBytes**: Efficient memory-optimized training.
- **Scikit-learn**: Metrics and auxiliary utilities.
- **WandB / TensorBoard**: Training monitoring and experiment tracking.

This modular design ensures flexibility, reproducibility, and scalability across both low-resource and high-capacity computing setups.

3.3 Software Implementation

For this experiment, we have used cloud computation software, Google Colab (Tesla T4: With 16GB of GDDR6 memory), and Kaggle (Tesla P100: The Tesla P100 GPU offers 16GB of HBM2 memory) for the computational task.

Chapter 4

Investigation/Experiment, Result, Analysis and Discussion

In this section, we present a detailed analysis of the experimental outcomes, focusing on three key phases: (1) base performance of student models without fine-tuning, (2) supervised fine-tuning of student models on the ethics dataset, and (3) reverse knowledge distillation (RKD) applied to larger teacher models.

For the training hyperparameters, we used:

No. of training epochs	10
Train Batch-size	16
Learning Rate	2e-4
Weight Decay	0.001
Optimizer	AdamW

Table 4.1: Hyperparameters used during finetuning Llama-2 (7B)

After fine-tuning the models, we have evaluated both the fine-tuned models on the binary ethics labels of our test set. The accuracy as well as the evaluation loss for both the fine-tuned models are reported below.

Table 4.2: Model Performance Comparison Across Stages

Model	Base (No Fine-tuning)		Fine-tuned SLM		After Reverse KD	
	Accuracy (%)	F1 Score	Accuracy (%)	F1 Score	Accuracy (%)	F1 Score
LLaMA 3.2B	67.4	0.68	81.2	0.82	-	-
Phi 3.5 Mini	64.9	0.66	79.8	0.80	-	-
Gemma 2 2B	65.7	0.67	80.4	0.81	-	-
LLaMA 2 7B	70.1	0.72	-	-	84.3	0.85
Gemma 2 9B	71.5	0.73	-	-	85.1	0.86

1. Baseline Performance of Small Language Models

We begin by evaluating the out-of-the-box performance of the small language models (SLMs) — LLaMA 3.2B, Phi 3.5 Mini Instruct, and Gemma 2 2B — on the ethics benchmark. These models were prompted using instruction-style queries without any task-specific tuning. As shown in Table 4.2, the average accuracy across the base models was around 66%–67%, with F1 scores ranging from 0.66 to 0.68. While these results indicate that SLMs possess some general ethical reasoning capabilities, they underperform on nuanced moral judgments and exhibit inconsistency across contextually ambiguous prompts.

2. Supervised Fine-Tuning of Student Models

To enhance task-specific alignment, we fine-tuned each student model using supervised learning on the ethics dataset. Fine-tuning was conducted using the Hugging Face Trainer API, employing a batch size of 16, a learning rate of $2e^{-5}$, and early stopping based on validation loss. Post-fine-tuning, all three SLMs demonstrated significant performance gains. For example, LLaMA 3.2B improved its accuracy from 67.4% to 81.2%, while the F1 score increased from 0.68 to 0.82. Similar improvements were observed with Phi 3.5 Mini and Gemma 2 2B. These results confirm that even compact models can effectively learn ethical reasoning when exposed to high-quality, human-annotated data.

3. Reverse Knowledge Distillation to Teacher Models

In the next phase, we applied our proposed **Reverse Knowledge Distillation (RKD)** technique to teach larger models such as LLaMA 2 7B and Gemma 2 9B using the knowledge distilled from the fine-tuned student models. Unlike traditional distillation where larger models teach smaller ones, here we reversed the gradient flow and optimized the teacher models to better emulate the high-accuracy behavior of the fine-tuned SLMs.

The distillation loss was defined using the KL-divergence between the soft logits of the student and teacher models. To avoid gradient vanishing or instability, we introduced a temperature scaling factor $\tau = 2.0$ and used a combined objective loss:

$$\mathcal{L}_{\text{total}} = \alpha \cdot \mathcal{L}_{\text{CE}} + (1 - \alpha) \cdot \mathcal{L}_{\text{KD}},$$

where $\mathcal{L}_{\text{CE}}$ is the original cross-entropy loss w.r.t. ground truth, and $\alpha = 0.5$ was used as a balancing coefficient.

After RKD, LLaMA 2 7B reached an accuracy of 84.3% and an F1 score of 0.85, while Gemma 2 9B achieved 85.1% accuracy and an F1 score of 0.86 — a substantial leap from their untuned baseline performance of around 71%. This validates that fine-tuned SLMs are capable of effectively transferring their distilled ethical reasoning to larger models that initially lacked contextual grounding in such moral tasks.

4. Additional Observations and Insights

- **Consistency and Robustness:** Fine-tuned SLMs displayed more stable and consistent predictions on edge-case ethical dilemmas compared to base models.
- **Inference Speed:** Despite being smaller, fine-tuned SLMs were able to deliver near-teacher-level accuracy with significantly lower computational cost, making them suitable for real-time ethical reasoning systems in constrained environments.
- **Error Analysis:** A qualitative review of incorrect predictions revealed that base models frequently struggled with implicit cultural cues, whereas fine-tuned models learned to handle context-sensitive prompts better.

Chapter 5

Impacts of the Project

5.1 Impact of this project on societal, health, safety, legal and cultural issues

The integration of ethical alignment into Small Language Models (SLMs) through the use of the ETHICS dataset and knowledge distillation from advanced Large Language Models (LLMs) addresses a wide array of real-world challenges. As these models are deployed in numerous settings—from customer service to education, healthcare, and digital communication—their ethical grounding ensures socially responsible and value-sensitive interactions. Societally, this project promotes equity, fairness, and non-discrimination in digital platforms. By training models to adhere to human-centric moral principles, we can reduce the harmful effects of algorithmic bias, which has been shown to disproportionately affect marginalized communities. These SLMs can serve as ethical filters in platforms that moderate content, provide legal information, or engage with users in sensitive conversations, ensuring responses are respectful and inclusive. In terms of legal and safety impacts, embedding ethical reasoning into SLMs helps developers meet emerging regulatory standards like the EU AI Act or U.S. AI Bill of Rights. These models are better positioned to avoid producing legally problematic content, such as discriminatory language or misinformation, reducing the liability for developers and organizations. Moreover, by anticipating ethical risks and minimizing harm, SLMs enhance user safety and public trust in AI. Health-related impacts are particularly significant. In healthcare, AI tools increasingly assist in symptom checking, mental health support, and patient education. Ethically aligned SLMs can respond with empathy, avoid stigmatizing language, and respect user privacy—making them more reliable and sensitive in these high-stakes environments. For instance, mental health chatbots built on such models can provide comfort and support while avoiding triggering or harmful language. Culturally, the alignment of AI systems with diverse ethical frameworks ensures they are sensitive to different traditions, languages, and moral viewpoints. The ETHICS dataset draws from universal ethical principles (e.g., fairness, virtue, justice) and promotes models that are aware of cultural nuance. This is essential

in multilingual, multicultural societies, where AI systems must avoid reinforcing stereotypes or ignoring minority perspectives.

5.1.1 Societal Impact

This project helps democratize access to ethical AI. Small Language Models are more accessible to institutions, startups, and public organizations with limited computing infrastructure. By ethically aligning these models, even low-resource settings can benefit from safe, intelligent systems. This ensures a more equitable distribution of AI technology across the globe. Furthermore, such models foster responsible digital communication in social media moderation, educational platforms, and news delivery, promoting a healthier information ecosystem.

5.1.2 Health Impact

Ethical SLMs are highly valuable in healthcare applications, especially where human-AI interaction is sensitive or personal. These models can assist in mental health therapy bots, virtual nurse assistants, and patient feedback systems by ensuring their language aligns with empathy, supportiveness, and respect. This avoids psychological harm, enhances patient satisfaction, and fosters trust in AI-enabled care solutions. The low-resource nature of these models also makes them suitable for use in rural or underserved healthcare environments, where advanced technology is often unavailable.

5.1.3 Cultural Impact

In an increasingly globalized digital world, the ability of AI to respect and adapt to cultural differences is crucial. This project supports the development of AI that understands context, respects local customs, and avoids culturally inappropriate or offensive behavior. The alignment process incorporates varied moral perspectives, such as deontological, utilitarian, and virtue ethics, enabling the creation of models that can interact effectively with diverse user bases. This is especially important for global corporations and multilingual platforms where a single ethical misstep can cause reputational damage.

5.2 Impact of this project on environment and sustainability

One of the most pressing concerns in modern AI development is energy consumption. Training and deploying LLMs requires vast computational resources, resulting in a large carbon footprint. This project mitigates that issue by focusing on knowledge distillation, a technique that transfers knowledge from large models to much smaller, more efficient ones. These SLMs require significantly less power to train and operate, making them more sustainable for long-term use. Moreover, the ETHICS dataset enhances the utility of these smaller models without needing to repeatedly retrain them from scratch, further conserving resources. The project aligns well with the goals of Green AI, which promotes resource-efficient methods without compromising performance. By making these ethical, compact models publicly available, the project also enables a broader, more sustainable deployment of AI technologies in education, governance, agriculture, and non-profit sectors—domains often left out of high-tech advancements due to limited budgets or infrastructure.

Chapter 6

Project Planning and Budget

Task Name	1st Term						2nd Term					
	24-Jul	24-Aug	25-Sep	26-Oct	27-Oct	28-Nov	29-Dec	30-Jan	30 Feb	1-Mar	2-Apr	2-May
Prepare Plan												
Research Literature												
Problem Finding												
Prepare Draft Budget												
Prepare & Submit Project Proposal												
Prepare & Submit 1st Term Progress Report												
Prepare & Present 1st Term Progress												
System Design												
Prototyping (Simulation)												
Prepare & Submit 2nd Term Progress Report												
Prepare & Present 2nd Term Progress												
Prototyping (Hardware)												
Testing & Evaluation												
Final Report												
Final Presentation												
Project Demonstration												

Figure 6.1: Gantt chart for project planning.

Component	Unit Price (in BDT)	Quantity	Total cost (in BDT)
Nvidia Geforce RTX 4090	2,78,000	1	2,78,000
Intel Core i7 14700K Raptor Lake Processor	32,000	1	32,000
Corsair H150 Liquid CPU Cooler	6,000	1	6,000
Corsair HX750 750W ATX Power Supply	7,500	1	7,500
Corsair 16GB DDR5 7200MHz Desktop RAM	14,500	4	58,000
ASRock Z790 Pro RS ATX Motherboard	30,000	1	30,000
Subtotal = 4,11,500 BDT (3734 USD)			

Figure 6.2: Budget for the project.

Chapter 7

Complex Engineering Problems and Activities

7.1 Complex Engineering Problems (CEP)

Attribute	Complex Engineering Problem (P)	Description
P1	Human-AI Alignment Knowledge	Aligning generative LLMs with human values using ethical-unethical labels from the ETHICS dataset.
P2	Combining LLM Reasoning with Human Logic	Integrating ethical reasoning and human logical thinking into the decision-making process of SLMs.
P3	Handling the Learning Process of LLMs	Understanding and applying the advanced architecture, quantization, and training flow of models like Llama 2 for ethical alignment.

Table 7.1: Complex Engineering Problems related to Human-AI Ethical Alignment

7.2 Complex Engineering Activities (CEA)

Activity	Description
Ethical Dataset Preparation	Prepared the ETHICS dataset to fine-tune models for ethical decision-making.
Bias Mitigation	Gathered diverse ethical reasoning samples from people of various domains to reduce dataset bias.
Framework & Libraries	Utilized PyTorch and Hugging Face Transformers library for implementation and model management.
Training Platforms	Used Google Colab and Kaggle Notebooks for model training and evaluation in a scalable, accessible way.
Resource-Efficient Training	Applied QLoRA to train large models in resource-constrained environments with high efficiency.

Table 7.2: Activities for Ethical Model Development and Training

Chapter 8

CONCLUSION

8.1 Summary

This project explores the ethical alignment of **Small Language Models (SLMs)** through a multi-stage methodology combining supervised fine-tuning and knowledge distillation from larger, pre-trained language models. The core motivation stems from the increasing demand for ethically grounded AI systems that are efficient enough to operate in resource-constrained environments. To that end, the study focuses on a set of lightweight yet expressive models, including LLaMA 3.2 1B, LLaMA 3.2 3B, Qwen 2.5 3B, and Phi 3.5 Mini 4B, aiming to imbue them with robust ethical reasoning capabilities.

The project leverages the **ETHICS** dataset—a comprehensive benchmark designed to test AI systems on nuanced moral reasoning and ethical dilemmas. As a first step, all selected SLMs underwent preprocessing and prompt engineering tailored to the dataset’s instruction-based format. These models were then fine-tuned using supervised learning to align their outputs with human-annotated moral decisions. Fine-tuning not only improved their performance but also enhanced their consistency in handling ambiguous or context-sensitive ethical queries.

To further enhance alignment, the study incorporated two complementary knowledge distillation approaches: *logit-based distillation*, where student models were trained to mimic the soft probability distributions of teacher models, and *response-based distillation*, which aligned student outputs with curated textual responses from larger models. Teacher models such as LLaMA 2 7B and Gemma 2 9B served as the ethical knowledge sources. A novel element of the methodology involved **reverse knowledge distillation**, where fine-tuned SLMs—after acquiring moral alignment—were used to further calibrate the responses of larger models, improving their behavior without full retraining.

Experimental results confirmed that small models, when fine-tuned and distilled effectively, can achieve moral reasoning performance close to that of their larger counterparts, while remaining significantly more efficient. These findings underscore the potential of using SLMs as ethically aligned alternatives for deployment in low-power or real-time AI applications. Ultimately, the project contributes to the broader goal of responsible AI development by enabling

scalable, ethical, and accessible language models suitable for diverse operational contexts.

8.2 Limitations

Despite the promising outcomes, several limitations were observed throughout the course of this research:

- **Limited Generalization Across Domains:** While the fine-tuned SLMs performed well on the ETHICS dataset, their performance may not generalize across other ethical or moral reasoning benchmarks such as Moral Scenarios or Social Chemistry. This raises concerns about domain-specific overfitting and the need for broader generalization.
- **Dataset Bias and Coverage:** The ETHICS dataset, despite its comprehensiveness, may not fully capture the diversity of ethical dilemmas encountered in real-world scenarios. Cultural, societal, and contextual biases embedded within the dataset can propagate through the fine-tuned models, potentially leading to skewed or non-universal moral judgments.
- **Model Capacity Constraints:** Small language models inherently possess fewer parameters, which limits their ability to capture complex abstract ethical reasoning patterns compared to larger models. While distillation helps, certain nuanced moral decisions may still be oversimplified due to limited model expressiveness.
- **Teacher-Student Misalignment:** The distillation process assumes that teacher models are fully ethically aligned, which may not always be the case. Inaccuracies, hallucinations, or misaligned reasoning in teacher responses could be transferred to students, especially in response-based distillation setups.
- **Reverse Distillation Scalability:** The concept of reverse distillation, though novel and experimentally effective, lacks rigorous theoretical grounding and scalability. The broader applicability of this technique remains uncertain across different architectures, tasks, or domains.
- **Evaluation Bottlenecks:** Ethical reasoning is inherently subjective and context-sensitive, making it difficult to evaluate model performance purely through quantitative metrics like accuracy and F1 score. More robust human-in-the-loop evaluations or alignment verification frameworks would provide better insight into real-world alignment.

- **Lack of Real-Time Testing:** The deployment of these ethically aligned SLMs in real-time or interactive systems was not explored in this study. As such, latency, robustness to adversarial prompts, and dynamic decision-making under uncertainty remain untested.
- **Resource-Intensive Distillation Pipeline:** Although the goal was to build efficient SLMs, the distillation process itself—especially when involving larger models—still required significant computational resources. This may contradict the initial objective of low-resource training pipelines in practical deployments.

8.3 Future Improvement

While this study provides valuable insights into the ethical alignment of Small Language Models (SLMs), several areas remain open for exploration in future work:

- **Cross-Domain Ethical Evaluation:** Expanding the evaluation to include a variety of ethical benchmarks, such as Moral Stories and Delphi, will help ensure that the models can generalize well across different domains and cultural contexts. This will also enable a deeper understanding of how ethical reasoning varies across different types of moral scenarios.
- **Multilingual Ethical Alignment:** Extending the approach to multilingual models is essential for global applicability. By incorporating more diverse languages, we can ensure that the ethical alignment of SLMs is robust across various cultural and linguistic contexts, addressing potential biases inherent in a single-language approach.
- **Human-in-the-Loop Feedback:** Incorporating real-time feedback from human evaluators during the training process could enhance the model’s ethical reasoning abilities. A more interactive framework, such as reinforcement learning from human feedback (RLHF), would allow for continuous improvement of ethical alignment based on dynamic human preferences.
- **Robustness to Adversarial Prompts:** Further research should investigate techniques to make ethically aligned models more resistant to adversarial or manipulative inputs. This is crucial to ensure that SLMs maintain ethical behavior even in the presence of misleading or harmful queries.

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